Practical 7: PHP Form Processing with Server-Side Validation and CSV/JSON File Storage Problem Definition Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

INDEX.PHP:-

```
<?php
$message = "";
$messageType = "";

$name = "";
$email = "";
$phone = "";
$subject = "";
$errors = [];

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    // Get and sanitize form data
    $name = trim($_POST["name"] ?? "");
    $email = trim($_POST["email"] ?? "");
```

```
    $phone = trim($_POST["phone"] ?? "");
    $subject = trim($_POST["subject"] ?? "");

    $name = htmlspecialchars($name, ENT_QUOTES, "UTF-8");
    $email = filter_var($email, FILTER_SANITIZE_EMAIL);
    $phone = htmlspecialchars($phone, ENT_QUOTES, "UTF-8");
    $subject = htmlspecialchars($subject, ENT_QUOTES, "UTF-8");

    // Server-side validation
    if ($name == "") {
        $errors[] = "Name is required.";
    }

    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $errors[] = "Enter a valid email address.";
    }

    if (!preg_match("/^[0-9]{10}$/", $phone)) {
        $errors[] = "Phone number must contain 10 digits.";
```

```
    }

    if ($subject == "") {
        $errors[] = "Message is required.";
    }

    // Store valid data in CSV file
    if (empty($errors)) {

        $file = fopen("data.csv", "a");

        if ($file !== false) {
            if (flock($file, LOCK_EX)) {
                if (fputcsv($file, [
                    $name, $email, $phone, $subject
                ]) !== false) {
                    $message = "Registration submitted
successfully!";
                    $messageType = "success";
                } else {
                    $message = "Error saving data.";
```

```
                $messageType = "error";
            }

            flock($file, LOCK_UN);
        } else {
            $message = "Unable to lock the data file.";
            $messageType = "error";
        }

        fclose($file);
    } else {
        $message = "Unable to open CSV file.";
        $messageType = "error";
    }
} else {
    $message = "Please correct the errors below.";
    $messageType = "error";
}
}
?>
```

```
<!DOCTYPE html>
<html>
<head>
    <title>Practical 7 - PHP Form</title>
    <style>
        body {
            font-family: Arial;
            background: #f2f2f2;
            margin: 30px;
        }

        .container {
            width: 400px;
            margin: auto;
            background: white;
            padding: 25px;
            border-radius: 10px;
        }

        input, textarea, button {
            width: 100%;
```

```
            padding: 10px;
            margin: 8px 0;
            box-sizing: border-box;
        }

        button {
            background: #087f5b;
            color: white;
            border: none;
            cursor: pointer;
        }

        .success {
            color: green;
        }

        .error {
            color: red;
        }
    </style>
</head>
```

```
<body>

<div class="container">
  <h2>Registration / Contact Form</h2>

  <?php if ($message != ""): ?>
    <p class="<?php echo $messageType; ?>">
      <?php echo htmlspecialchars($message); ?>
    </p>
  <?php endif; ?>

  <?php if (!empty($errors)): ?>
    <ul class="error">
      <?php foreach ($errors as $error): ?>
        <li><?php echo htmlspecialchars($error); ?></li>
      <?php endforeach; ?>
    </ul>
  <?php endif; ?>

  <form method="POST" action="">
    <label>Name:</label>
```

```
        <input type="text" name="name"
            value="<?php echo $name; ?>" required>

        <label>Email:</label>
        <input type="email" name="email"
            value="<?php echo htmlspecialchars($email); ?>"
            required>

        <label>Phone:</label>
        <input type="text" name="phone"
            value="<?php echo $phone; ?>"
            maxlength="10" required>

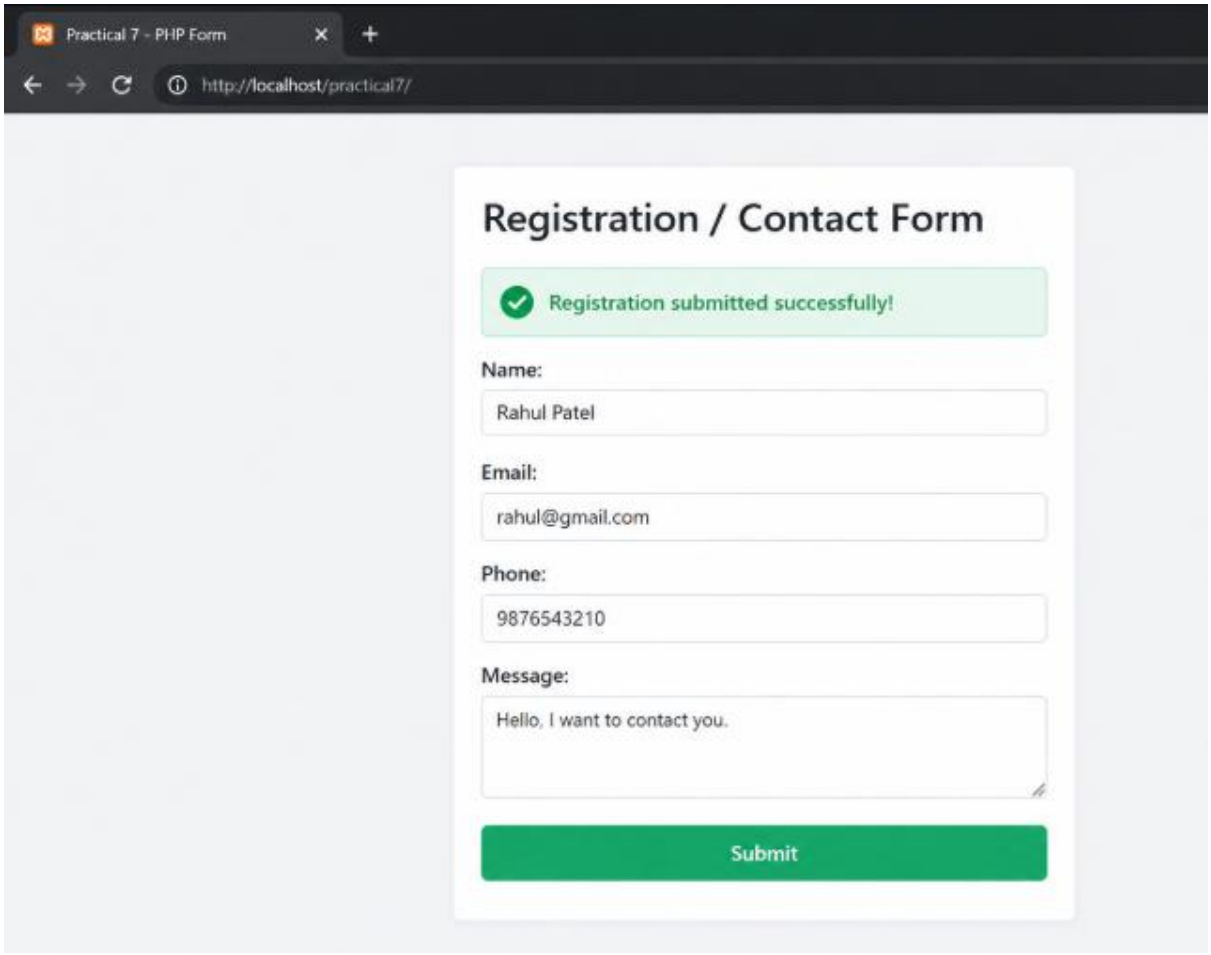
        <label>Message:</label>
        <textarea name="subject" required><?php
           echo $subject;
        ?></textarea>

        <button type="submit">Submit</button>
    </form>
</div>
```

```
</body>

</html>
```

OUTPUT:-





Key Questions / Analysis / Interpretation

1. Is the form submitted using POST?

Yes, the form uses method="POST" to send data to the PHP server.

2. Are inputs validated and sanitized server side?

Yes, PHP validates the name, email, phone number and message, and sanitizes the inputs before storing them.

3. Is file writing handled safely?

Yes, fopen() and fputcsv() are used to store data in a CSV file, and flock() helps prevent simultaneous writes from interfering with each other.

4. Are success and error responses displayed clearly?

Yes, clear success messages are displayed when data is stored, and error messages are displayed when validation or file writing fails.

Tools/Technology to be Used

VSCODE , PHP, XAMPP/WAMP/LAMP, browser dev tools